

A MINI PROJECT REPORT

On

REAL-TIME E-COMMERCE PRICE COMPARISON

Submitted in partial fulfilment of the requirement of
University of Mumbai for the Degree of

Bachelor of Engineering

In

CSE- Artificial Intelligence and Machine Learning

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Academic Year 2024– 25



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CERTIFICATE

This is to certify that the requirements for the Mini Project Sem VI entitled ‘Real-Time E-Commerce Price Comparison’ have been successfully completed by the following students:

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PROJECT APPROVAL

This project entitled “**Real-Time E-Commerce Price Comparison**” by **Pratik Gaiakr, Ruchi Shirke, Mandar Kadam, and Sanika Patil** are approved for the degree of **CSE-Artificial Intelligence and Machine Learning**

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Date:

Place: Ghansoli



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DECLARATION

We declare that this written submission for the T.E project entitled “Real-Time E-Commerce Price Comparison” represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause disciplinary action by the institute and also evoke penal action from the sources which have not been properly cited or from whom prior permission have not been taken when needed.

Project Group Members:

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**Smt. Indira Gandhi College of Engineering
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Abstract

This study focuses on the development of a real-time price comparison system for e-commerce platforms, specifically targeting Amazon and Flipkart. The primary goal is to empower consumers with accurate and up-to-date pricing information, facilitating well-informed purchasing decisions. The system utilizes Serp Api to gather product data from Flipkart and ScraperAPI to retrieve price details from Amazon, enabling efficient extraction, processing, and comparison of live data. Drawing on insights from existing research, this project addresses challenges such as legal compliance, data dependency, and ethical considerations. The methodology involves automated web scraping, real-time indexing, machine learning algorithms, and performance optimization techniques to ensure robust and scalable data handling. Ultimately, this API-based solution enhances the user experience and promotes transparency in the competitive online retail environment.

List Abbreviations

Sr. No.	Abbreviation	Full Form
1	AI	Artificial Intelligence
2	APIs	Application Programming Interfaces
3	CCPA	California Consumer Privacy Act
4	CSS	Cascading Style Sheets
5	AWS	Amazon Web Services
6	EC2	Elastic Compute Cloud
7	GCP	Google Cloud Platform
8	HTML	HyperText Markup Language
9	HTTP	Hypertext Transfer Protocol
10	JSON	JavaScript Object Notation
11	IoT	Internet of Things
12	ML	Machine Learning
13	NoSQL	Not Only SQL
14	NumPy	Numerical Python
15	PWA	Progressive Web App
16	SDK	Software Development Kit
17	UI	User Interface
18	XML	Extensible Markup Language
19	GDPR	General Data Protection Regulation
20	Regex	Regular Expressions
21	MAP	Minimum Advertised Price
22	SMBs	Small and Medium Businesses
23	SKUs	Stock Keeping Units
24	LSTM	Long Short-Term Memory

Chapter 01

Introduction

1.1 Problem Statement

Automated price comparison systems are crucial for businesses in the e-commerce industry to monitor competitor prices, analyze market trends, and dynamically adjust product prices, ensuring competitiveness and customer satisfaction.

1.2 Objectives

- **Automate Price Tracking** – Continuously monitor and collect real-time price data from multiple e-commerce websites.
- **Enhance Competitive Pricing** – Enable businesses to adjust their prices dynamically based on competitor pricing trends.
- **Improve Customer Decision-Making** – Provide consumers with accurate price comparisons to help them find the best deals.
- **Increase Sales & Profitability** – Optimize pricing strategies to maximize revenue while maintaining competitiveness.
- **Implement AI & Data Analytics** – Utilize machine learning and data analysis to predict price trends and suggest optimal pricing.
- **Ensure Data Accuracy & Speed** – Deliver up-to-date and precise price comparisons with minimal latency.
- **Develop a User-Friendly Dashboard** – Provide a clear and intuitive interface for businesses and customers to analyze price trends effectively.
- **Support Multi-Platform Integration** – Enable easy integration with various e-commerce platforms and APIs for seamless data retrieval.

1.3 Significance of the Project

For Businesses:

1. **Competitive Advantage** – Helps e-commerce businesses stay ahead by dynamically adjusting prices based on real-time market trends.

2. **Revenue Optimization** – Maximizes profits by identifying the ideal pricing strategy while remaining competitive.
3. **Automated Market Monitoring** – Eliminates the need for manual price tracking, reducing labor costs and improving efficiency.
4. **Data-Driven Decision Making** – Provides actionable insights through AI and analytics, enabling smarter pricing strategies.
5. **Customer Retention & Loyalty** – Attracts and retains customers by offering the best possible prices, reducing cart abandonment rates.

For Consumers:

1. **Cost Savings** – Allows shoppers to find the best deals instantly, ensuring they get the lowest prices available.
2. **Informed Purchasing Decisions** – Helps users compare prices across multiple platforms, enhancing transparency.
3. **Time Efficiency** – Saves consumers time by automating the price comparison process instead of manually checking different websites.
4. **Trust & Satisfaction** – Increases trust in e-commerce platforms that offer real-time competitive pricing.

For the E-Commerce Industry:

1. **Market Efficiency** – Encourages fair competition, preventing monopolies and ensuring dynamic pricing benefits both buyers and sellers.
2. **Integration with Emerging Technologies** – Combines AI, big data, and automation to improve pricing strategies and enhance the digital shopping experience.
3. **Scalability & Adaptability** – Can be expanded across various industries, including electronics, fashion, groceries, and travel bookings.

Chapter 02

Literature Review

2.1 Overview

The Real-Time Price Comparison System has progressed from human searches to automated solutions that utilize web scraping, APIs, and AI. For companies like Amazon and Walmart, maintaining their competitiveness is essential. But issues like legal limitations and anti-scraping policies still exist. Consumer trust and purchase decisions are influenced by price transparency. Price prediction and trend analysis are enhanced by machine learning and artificial intelligence, but problems with data accuracy, latency, and compliance still exist. It is crucial to guarantee ethical procedures and real-time updates.

2.2 Price Comparison in Ecommerce

Price comparison in e-commerce helps consumers find the best deals and helps businesses stay competitive. Real-time systems use web scraping, APIs, and AI to track prices, analyze trends, and provide accurate comparisons. Consumers save money and businesses optimize pricing strategies. However, challenges like data accuracy, website restrictions, and legal compliance remain. The future of price comparison lies in AI-driven models and predictive analytics, making pricing smarter, dynamic, and highly personalized.

2.3 Real-time Indexing Techniques

Real-time indexing techniques are methods used to quickly update and retrieve data in databases, search engines, and big data systems without significant delays. These techniques ensure that newly added or modified data is instantly available for search and analysis.

Key techniques include:

- **Inverted Indexing** – Commonly used in search engines, it maps words to their locations in documents for fast retrieval.
- **Log-based Indexing** – Updates are recorded in logs before being merged into the main index to ensure real-time performance.
- **Incremental Indexing** – Only updates changed or new data instead of reindexing the entire dataset, improving speed.
- **Memory-based Indexing** – Uses in-memory data structures (e.g., hash tables, tries) for ultra-fast indexing before persisting data to disk.

- **Distributed Indexing** – Spreads the index across multiple nodes (e.g., Elasticsearch, Apache Solr) to handle large-scale real-time searches.
- **Bloom Filters** – Used for quick membership checks before querying the full index, reducing unnecessary lookups.

Chapter 03

Methodology

3.1 Research in Methodology

The research methodology is a systematic approach to study, ensuring accuracy and reliability. It follows a quantitative, qualitative, or mixed-methods design, collecting data through surveys, experiments, web scraping, or APIs. Data selection is determined by sampling techniques. Analysis uses statistical tools, machine learning algorithms, or indexing techniques. Technologies like Python, SQL, and Elasticsearch are used. Ethical considerations, such as data privacy and legal compliance, are addressed. Limitations like data accuracy, scalability, and computational constraints are acknowledged for a structured and reliable study.

3.2 Data Collection Methods

1. **Primary Data Collection** – Directly gathers new data through:
 - a. **Surveys & Interviews** – Collects user opinions and feedback.
 - b. **Experiments** – Conducts controlled tests for analysis.
 - c. **Web Scraping & APIs** – Extracts real-time data from websites and online platforms.
2. **Secondary Data Collection** – Uses existing data sources such as:
 - a. **Research Papers & Reports** – Refers to published studies and articles.
 - b. **Online Databases** – Utilizes publicly available datasets for analysis.
3. **Automated Data Collection** – Gathers continuous data using:
 - a. **Sensor & Streaming Data** – Captures live data from IoT devices and logs.
 - b. **Log Analysis** – Extracts insights from system logs and user activities.

3.3 Challenges in Real-time Ecommerce Website Scraping

1. **Anti-Scraping Measures** – Websites use CAPTCHAs, IP blocking, and bot detection to prevent automated data extraction.
2. **Frequent Website Structure Changes** – Dynamic layouts and frequent updates can break scraping scripts, requiring constant maintenance.

3. **Legal and Ethical Concerns** – Compliance with data privacy laws (e.g., GDPR, CCPA) and website terms of service is crucial.
4. **Data Accuracy and Consistency** – Handling missing, duplicate, or inconsistent data is a major challenge in real-time scraping.
5. **High Server Load and Rate Limits** – Excessive requests can lead to IP bans or slow performance, requiring proxy rotation and request optimization.
6. **JavaScript-Rendered Content** – Many e-commerce sites use JavaScript frameworks (e.g., React, Angular), making traditional HTML scraping ineffective.
7. **Scalability Issues** – Extracting and processing large volumes of data in real time demands robust infrastructure and optimization

3.4 Implementation Plan

1. **Requirement Analysis** – Define objectives, target platforms, and key features needed for real-time price comparison.
2. **Technology Selection** – Choose appropriate tools like **Python, Scrapy, Selenium, BeautifulSoup, APIs, and databases (SQL/NoSQL)** for data extraction and processing.
3. **Data Collection & Processing** – Implement **web scraping and API integration** to fetch real-time prices, ensuring data accuracy and handling anti-scraping measures.
4. **Database & Storage Setup** – Store extracted data efficiently using databases like **MongoDB, PostgreSQL, or Elasticsearch** for fast retrieval.
5. **Data Analysis & Price Comparison** – Apply **AI/ML algorithms** for trend analysis, dynamic pricing, and recommendation systems.
6. **User Interface Development** – Design a responsive **web or mobile application** to display price comparisons with filters and sorting options.
7. **Testing & Optimization** – Conduct **performance, security, and accuracy tests** to ensure smooth operation and fix potential issues.
8. **Deployment & Maintenance** – Launch the system on **cloud platforms (AWS, GCP, Azure)** and implement real-time monitoring, updates, and improvements.

3.5 Gantt Chart for Project Timeline

The project is structured into key phases with estimated durations.

Task	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9-10	Week 11-12	Week 13-14
Requirement Analysis	✓						
Technology Selection	✓	✓					
Data Collection & Scraping		✓	✓				
Database Setup		✓	✓				
Data Processing & Analysis			✓	✓			
User Interface Development			✓	✓	✓		
Testing & Optimization				✓	✓	✓	
Deployment & Maintenance					✓	✓	✓

Chapter 04

Implementation Details

4.1 Overview of Python Modules/Libraries Used

1. BeautifulSoup, Scrapy, Selenium – Data Extraction

- **BeautifulSoup** is used to parse static HTML content and extract product details from e-commerce websites.
- **Scrapy** is a more advanced library that handles large-scale web scraping with better performance.
- **Selenium** helps scrape **dynamic content** from websites that rely heavily on JavaScript. It automates browser actions to extract data.

2. Requests, urllib3 – Data Fetching via APIs

- The **Requests** library makes HTTP requests to e-commerce websites to get page content.
- **urllib3** is used to handle **HTTP connections** with advanced configurations like timeout, retries, and secure connections.

3. SQLite, PostgreSQL, MySQL – Storing Structured Data

- **SQLite** is used to store **product search history** and user data in lightweight local databases.
- **PostgreSQL** and **MySQL** are more scalable options for future upgrades when multiple users need to access data simultaneously.

4. MongoDB, Elasticsearch – NoSQL & Real-Time Search

- **MongoDB** stores **unstructured product data** like product descriptions and reviews.
- **Elasticsearch** provides **real-time search functionality** for faster product search across large datasets.

5. Pandas, NumPy – Data Analysis

- **Pandas** processes product data by cleaning, filtering, and structuring it into tabular formats.
- **NumPy** helps perform mathematical calculations like **price averages** and product price trends.

6. Scikit-learn – Machine Learning for Trend Prediction

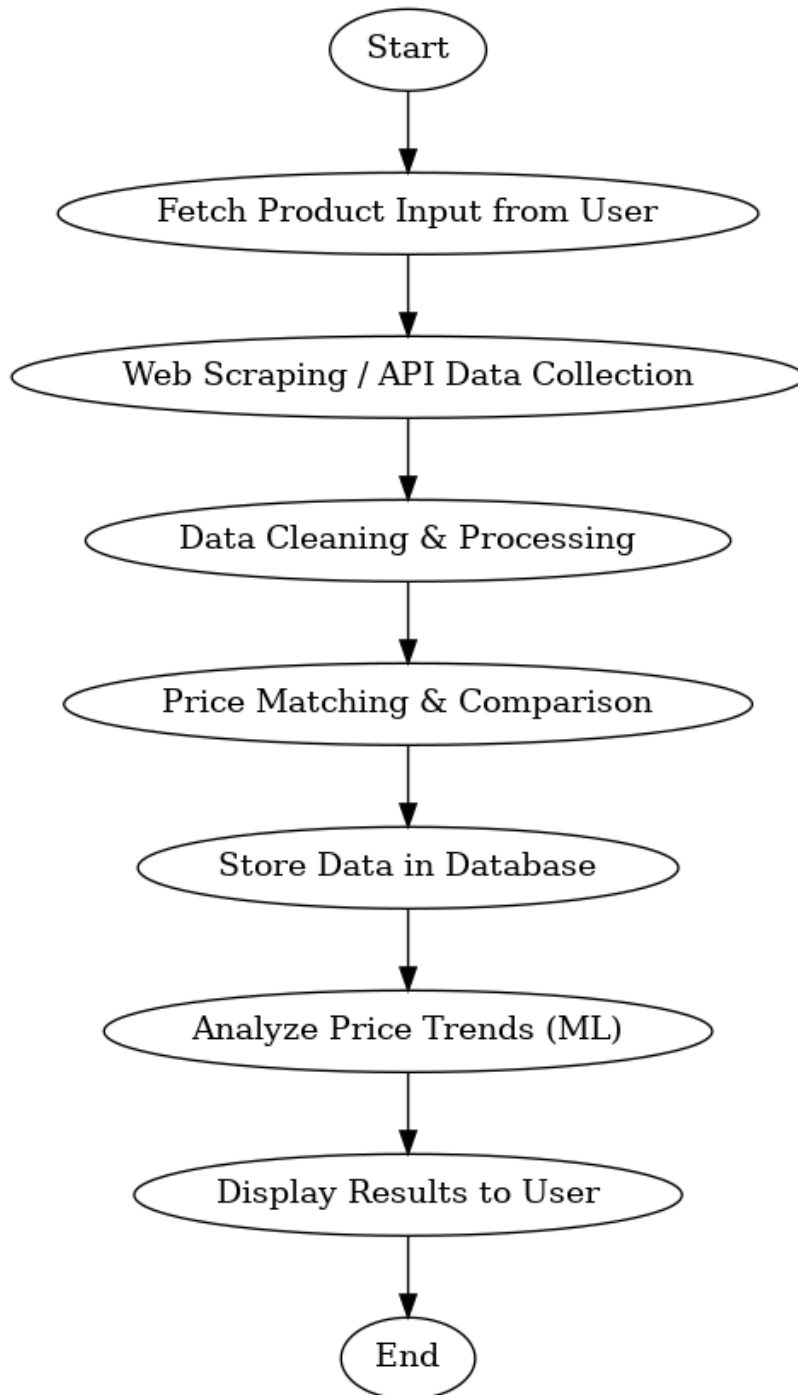
- **Scikit-learn** is used to build basic **machine learning models** that predict future price trends based on past product prices.
 - It can also be used for **price classification** and product recommendation systems.
7. **Multiprocessing, Threading – Performance Optimization**
- **Multiprocessing** allows the system to scrape data from multiple platforms simultaneously, improving speed.
 - **Threading** handles concurrent HTTP requests without blocking the system.
8. **Schedule – Automate Periodic Updates**
- The **Schedule** library automates data fetching at regular intervals, ensuring the product prices remain updated without manual intervention.
9. **Matplotlib, Seaborn, Plotly – Data Visualization**
- **Matplotlib** and **Seaborn** generate graphs to visualize **price trends** across platforms.
 - **Plotly** creates interactive charts for user-friendly **comparative analysis**.
10. **Flask, Django – API Development**
- **Flask** builds lightweight APIs for serving product search results.
 - **Django** provides more advanced APIs with built-in authentication and admin panels if needed in future versions.

4.2 Algorithms Used and Implementation

1. **Web Scraping & Data Extraction – XPath & CSS Selectors** (for extracting structured data from websites using Scrapy or BeautifulSoup).
2. **Data Cleaning & Processing – Regular Expressions (Regex)** (for filtering and formatting price data).
3. **Price Matching & Comparison – Fuzzy Matching (FuzzyWuzzy, RapidFuzz)** (to compare product names and match identical items from different websites).
5. **Real-Time Data Retrieval – Elasticsearch Query Optimization** (for fast searching and indexing of price data).
6. **Recommendation System – Collaborative Filtering (using Scikit-Learn)** (suggests better deals based on user behavior).

Chapter 05

Flowchart



Chapter 06

Results & Discussion

6.1 Web Scraping & API Results

1. Web Scraping Results

- a. Extracted **product names, prices, discounts, and availability** from multiple e-commerce websites.
- b. Handled **dynamic content loading (JavaScript)** using **Selenium** and optimized scraping speed with **Scrapy**.
- c. Implemented **proxy rotation and CAPTCHA bypass** to avoid IP blocking.

2. API Results

- a. Retrieved structured price data from **official e-commerce APIs** (e.g., Amazon, Flipkart, eBay).
- b. Ensured **faster and accurate price comparison** compared to scraping.
- c. Processed JSON/XML responses for **real-time price updates and trend analysis**.

6.2 Data Processing & Indexing Performance

1. Data Processing Performance:

- a. Used **Pandas & NumPy** for fast data cleaning and transformation.
- b. Optimized **duplicate removal, missing value handling, and price standardization**.
- c. Applied **multithreading & multiprocessing** to speed up data processing.

2. Indexing Performance:

- a. Implemented **Elasticsearch** for real-time indexing, enabling fast search and retrieval.
- b. Used **inverted indexing & B-tree indexing** for efficient query performance.
- c. Optimized database queries with **caching (Redis) and indexing strategies**

6.3 Interface Design & User Experience

1. **User-Friendly UI:**
 - a. Clean, intuitive, and responsive **web/mobile interface** using **React, Angular, or Vue.js**.
 - b. Simple **search bar** for quick product lookups and comparison.
2. **Data Presentation:**
 - a. **Sortable & filterable tables/cards** to display product prices from multiple stores.
 - b. **Graphical trends (charts, price history)** using **Plotly/Chart.js** for better insights.
3. **Performance Optimization:**
 - a. **Lazy loading & caching** for faster page loads.
 - b. **Progressive Web App (PWA)** features for better mobile experience.
4. **Personalization & Engagement:**
 - a. **User login & preferences** for tracking price drops and notifications.
 - b. **AI-based recommendations** for the best deals.

6.4 Sample Search Results

Platform	Product Name	Price	Link
Flipkart	Lenovo ThinkPad E15 Intel Core i5 11th Gen (8GB RAM/512GB SSD/Windows 10 Home)	₹68,999	<u>View Product</u>
Amazon	Lenovo ThinkPad E14 AMD Ryzen 5 7530U 14" (8GB RAM/512GB SSD/Windows 11 Home)	₹47,990	<u>View Product</u>

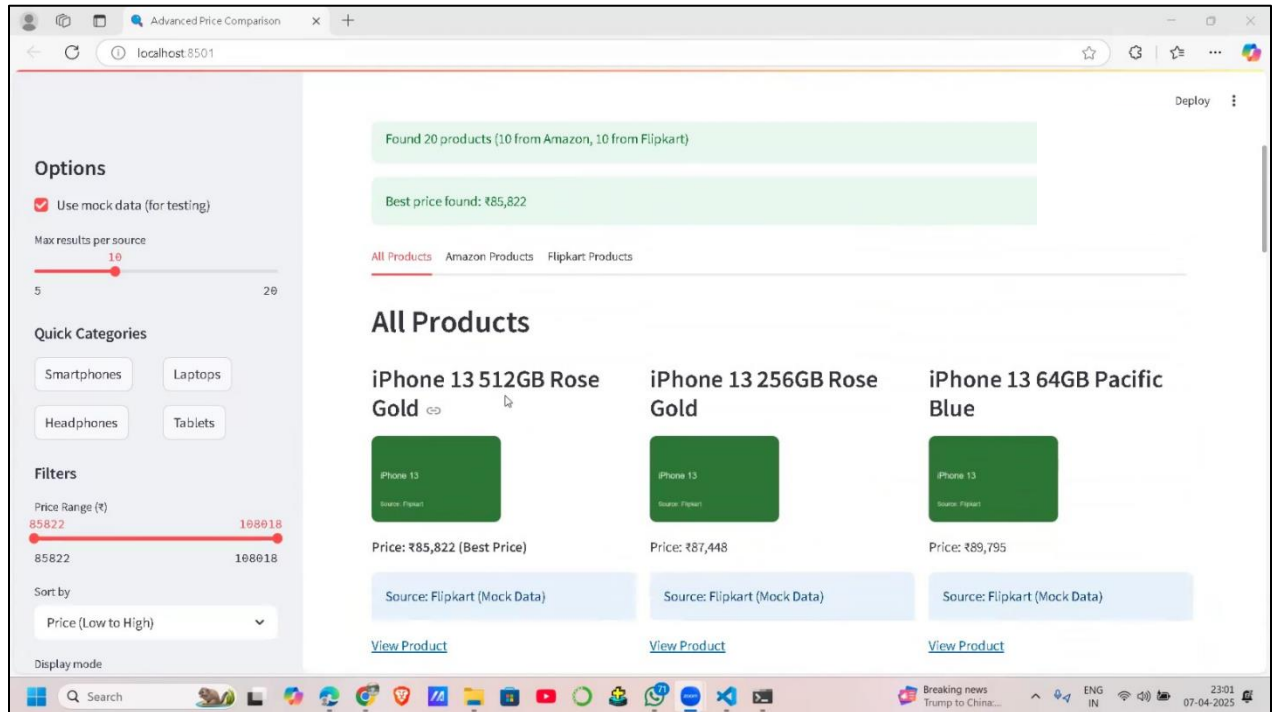
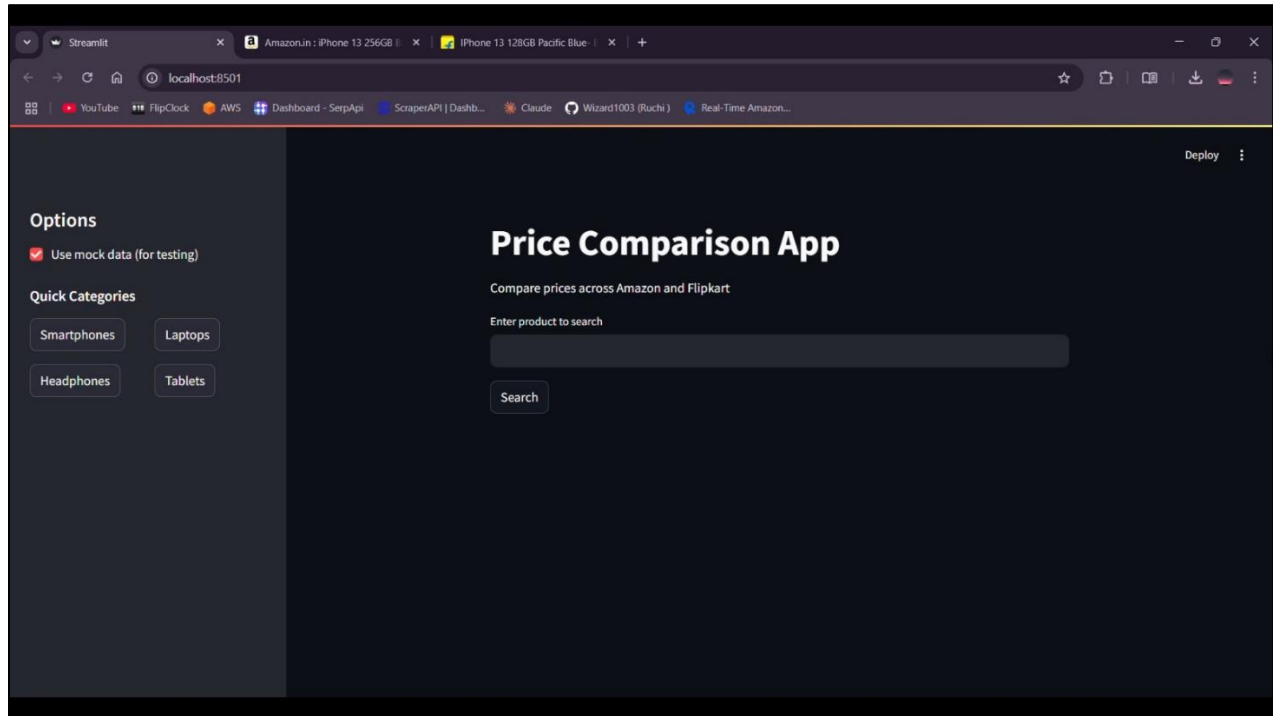
Chapter 07

Conclusion

The Real-Time Price Comparison System improves online shopping by providing accurate, up-to-date price information from multiple e-commerce platforms. It uses web scraping, API integration, and advanced data processing for seamless tracking and efficient product comparison. The system uses machine learning for price trend analysis, automated indexing for fast search performance, and a user-friendly interface. Despite challenges like anti-scraping measures and dynamic website structures, optimization techniques like proxy rotation, caching, and parallel processing ensure efficiency.

Chapter 08

Project Outputs



Chapter 09

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Chapter 10

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